

DRO-FAIR: Adversarial Fairness Attacks (Week 2)

May 29, 2026 - Progress Report for Madam

Executive Summary

This week we implemented gradient-based PGD attacks targeting fairness metrics (DP, IF, Joint) and tested DRO-FAIR performance on 270 experiments across 3 datasets (Adult, Credit, LSAC).

Key Results

- Credit IF-Attack $\alpha=0.2$: DRO reduces DP by 64.5% ($p=0.031$) ***
- Credit IF-Attack $\alpha=0.3$: DRO reduces DP by 97.5% ($p=0.031$) ***
- LSAC IF-Attack $\alpha=0.3$: DRO reduces DP by 96.2% ($p=0.031$) ***
- Adult DP-Attack: DRO shows higher DP than Naive (known feedback loop)
- Combined attacks are weaker than single-metric attacks (budget split)

Method

FairnessTargetedPGD computes the exact gradient of the fairness metric w.r.t. each sample label, then flips the top- α samples that maximize unfairness. Three attack modes: DP-only, IF-only (k-NN based), and Joint (weighted sum).

Experiments

Total: 270 experiments = 3 datasets x 3 alphas x 5 seeds x 3 attacks x 2 methods. Fast mode: epochs=30, K_inner=5.

UTKFace Status

Pipeline ready: ResNet18 feature extractor and experiment runner implemented. Smoke test passes. Waiting on GPU server access.

Deliverables

- src/corruption/adversarial.py - FairnessTargetedPGD
- figures/fig8_fairness_pgd_comparison.png - Bar chart
- figures/fig9_fairness_pgd_curves.png - Line chart
- results/fairness_pgd_wilcoxon.csv - Statistical tests
- docs/ADVERSARIAL_FAIRNESS_REPORT.md - Full report